

Systematic Analysis of Weight Quantization Thresholds in Convolutional Neural Networks: A Dataset-Dependent Study

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ABSTRACT

While neural network quantization significantly reduces memory and computational overhead, the reciprocity between weight bitwidth and accuracy across varying dataset complexities remains under-explored. This study systematically analyzes weight quantization thresholds by evaluation through Convolutional Neural Networks (CNNs) trained with 1-bit, 4-bit, 8-bit, and 32-bit weights on the MNIST and CIFAR-10 datasets. Our empirical results reveal that quantization tolerance is extremely sensitive to dataset complexity. Simple tasks like MNIST sustain peak performance (99.10%) even under 1-bit constraints, whereas CIFAR-10 exhibits a critical performance cliff between 4-bit and 8-bit precision. We identified 8-bit quantization as the optimal “sweet spot” for complex tasks, offering a 4 times compression ratio with a marginal 4.15% accuracy loss. By isolating weight bitwidth from activation and gradient precision, we provide a controlled characterization of its independent impact on generalization. These findings offer a practical framework for matching bitwidth to task complexity, enabling extreme compression in resource-constrained environments without sacrificing performance.

I. INTRODUCTION

Deep neural networks have achieved unparalleled efficacy across a broad spectrum of computational domains, most notably in computer vision, natural language processing, (Krizhevsky et al., 2012) and speech recognition (Hinton et al., 2021a). This proliferated utility has positioned connectionist architectures as the preeminent framework for high-dimensional pattern recognition, with contemporary models consistently surmounting established human-level benchmarks across standardized datasets.

Despite their efficacy, the deployment of modern neural architectures is frequently constrained by their immense computation and memory footprints. The expansion of information of high-capacity models, often comprising billions of parameters, renders them unsuitable for hardware-limited substrates such as mobile devices. In these contexts, restricted memory bandwidth and power budgets represent the most critical challenges that prevent this technology from being accepted or used. Beyond hardware limitations, the grand size of these models terribly complicate deployment in applications where latency reduction is imperative and immediate inference is crucial.

To mitigate these computational overheads, various model compression techniques have been proposed including structural pruning [Han et al., 2015], knowledge distillation [Hinton et al., 2015], and low-rank factorization [Denton et al., 2014]. Among

these methodologies, weight quantization (the process of mapping network parameters to a discretized, reduced-precision numerical space) presents interesting synergy between efficiency and performance. By facilitating substantial reductions in memory footprint while maintaining smooth, continuous and differentiable operations for gradient-based optimization, quantization has emerged as a primary strategy for edge-side acceleration. Recent works indicate a surprising degree of precision resilience in neural architectures; notably, binary neural networks have demonstrated the capacity to retain competitive accuracy even when weights are constrained to a 1-bit representation [Courbariaux & Bengio, 2016; Rastegari et al., 2016]

However, the systematic mapping of quantization levels to accuracy loss remains significantly under-explored. Significant ambiguity persists regarding: (1) the numerical inflection points beyond which model utility is compromised; (2) how these requirements scale with dataset complexity; and (3) the existence of generalized predictive model for the compression-accuracy trade off. Unfortunately existing works often lack a strict, controlled research that validates a model by removing or modifying specific components, as the quantization of multiple network components prevents isolated assessment of weight impact.

This paper addresses these gaps through empirical investigation of weight quantization across a spectrum of precision levels on datasets of varying complexity. By isolating weight quantization while maintaining full precision for activations and gradients, we provide controlled analysis of weight bitwidth’s independent impact on model performance.

This paper makes the following contributions:

1. We show that quantization thresholds are not universal but are contingent upon dataset complexity, contrasting compression viability of MNIST against a more complex and stricter CIFAR-10.
2. We pinpoint the 4-bit to 8-bit range as the decisive interval for complex classification tasks, establishing 8-bit as the gold standard for balancing model footprint and predictive accuracy.
3. We demonstrate the non-linear nature of the precision-accuracy trade-off, providing evidence for a bounded worst-case loss that informs predictable deployment planning.

4. Translate experimental findings into actionable guidelines for selecting bitwidth configurations based on specific resource constraints and task-inherent difficulty.

II. METHODS

A. Weight Quantization Methodology

We isolate the effect of weight quantization by maintaining full 32-bit precision for activations and gradients while varying weight bitwidth across four levels: 1-bit, 4-bit, 8-bit and 32-bit (baseline). This weight-only configurations enables controlled analysis of quantization impact without confounding effects from activation or gradient quantization.

For k -bit weight quantization ($k > 1$), we adopt the quantization function from Zhou et al [14]

$$W_q = 2 \cdot Q_k((\tanh(W))/(2 \cdot \max|\tanh(W)|) + 0.5) - 1 \quad (1)$$

where W represents full-precision weights and Q_k is the k -bit quantizer:

$$Q_k(x) = \text{round}(x \cdot (2^{k-1})) / (2^{k-1}) \quad (2)$$

For binary (1-bit) weights, we use:

$$W_q = \text{sign}(W) \cdot \text{mean}(|W|) \quad (3)$$

Since quantization is non-differentiable, we use the straight-through estimator [2, 14] for gradient backpropagation, approximating $\partial L / \partial W \approx \partial L / \partial W_q$. This allows gradient-based optimization despite discrete weight values.

B. Network Architecture and Training

We additionally use a convolutional neural network with two convolutional layers ($3 \rightarrow 32 \rightarrow 64$ filters, 3×3 kernels) followed by max pooling and two fully-connected layers (128 hidden units, 10 output classes). The network contains approximately 545,000 parameters, all subject to the target quantization level.

Training follows a dual-precision procedure: weights are stored in full precision (32-bit), quantized before each forward pass, and updated in full precision after backpropagation. This prevents loss in gradient information while enabling low-bitwidth inference. We use the Adam optimizer [3] with learning rate 0.001, batch size 128, and cross-entropy loss. Models are trained from random initialization (Kaiming normal [4]) for 5 epochs on both datasets.

C. Datasets & Evaluation

We evaluate two datasets of varying complexity:

MNIST: 60,000 training and 10,000 test images of handwritten digits (28x28 grayscale, resized to 32x32 RGB), representing a simple classification task with high class separability.

CIFAR-10: 50,000 training and 10,000 test images across 10 natural object categories (32x32 RGB) representing a complex classification problem with lower separability and higher-class variance.

For each bitwidth configuration (1, 4, 8, 32 bits) we will measure the following:

- Test accuracy (primary metric)

- Model size: (total parameters $\times$ bitwidth) / (8 $\times$ 1024²) MB
- Compression ratio: $\text{Size}_{32\text{-bit}} / \text{Size}_{k\text{-bit}}$
- Accuracy retention: $\text{Acc}_{k\text{-bit}} / \text{Acc}_{32\text{-bit}} \times 100\%$

We will report on the best test accuracy achieved during training.

D. Training Algorithm

Algorithm 1 summarizes our training procedure, which implements dual-precision weight management for quantized neural networks.

Algorithm 1: Quantized Weight Training (Weight-Only Configuration)

Input: Training data D , learning rate η , target bitwidth k

Output: Trained model with quantized weights

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1: Initialize full-precision weights  $W \in \mathbb{R}$  (Kaiming normal)
2: for each epoch do
3:   for each minibatch  $(x, y)$  in  $D$  do
4:     // Forward pass with quantized weights
5:      $W_q \leftarrow \text{Quantize}(W, k)$  (Eq. 1, 2, or 3)
6:      $\hat{y} \leftarrow \text{Network}(x; W_q)$  (using  $W_q$ )
7:      $L \leftarrow \text{CrossEntropy}(\hat{y}, y)$ 
8:
9:     // Backward pass with straight-through estimator
10:     $\nabla W_q \leftarrow \partial L / \partial W_q$  (standard backprop)
11:     $\nabla W \leftarrow \nabla W_q$  (straight-through)
12:
13:    // Update full-precision weights
14:     $W \leftarrow \text{Adam}(W, \nabla W, \eta)$  (in full precision)
15:  end for
16: end for
17: return  $W_q \leftarrow \text{Quantize}(W, k)$ 
    (final quantized model)

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Our training algorithm is inspired by the DoReFa-Net framework proposed by Zhou et al. [14], but differs in scope and application. DoReFa-Net introduces a quantization strategy that simultaneously reduces precision for three network components: weights (W), activations (A), and gradients (G). Their complete algorithm quantizes all three during both forward and backward passes, enabling bit-wise convolution operations throughout training to accelerate computation on specialized hardware.

Our implementation, however, takes only the weight quantization component of DoReFa-Net (specifically, their Equation 9 for k -bit weights and Equation 7 for binary weights), while maintaining full 32-bit precision for activations and gradients. Where DoReFa-Net's typical configuration might be $W=1, A=2, G=6$ (1-bit weights, 2-bit activations, 6-bit gradients), our configurations are $W \in \{1, 4, 8, 32\}, A=32, G=32$ (variable weight precision, full-precision activations and gradients).

This simplification serves the research objective: to isolate and quantify the independent effect of weight quantization on model performance. By holding activations and gradients constant at full precision, we eliminate confounding variables and can directly attribute any observed accuracy degradation to weight bitwidth reduction alone. DoReFa-Net's joint quantization approach, while more computationally efficient, does not permit such controlled analysis since performance changes could result from interactions between weight, activation, and gradient quantization.

We also have DoReFa-Net's dual-precision training strategy (maintaining full-precision weight accumulators while using quantized weights for forward passes) and their straight-through estimator for gradient propagation. These components are necessary for stable training and are independent of whether activations and gradients are also quantized.

III. RESULTS

	Dataset	Bitwidth (w)	Best Test Acc (%)	Model Size (MB)	Compression Ratio
3	CIFAR10	32	70.45	2.08	1.0
2	CIFAR10	8	66.81	0.52	4.0
1	CIFAR10	4	63.71	0.26	8.0
0	CIFAR10	1	62.71	0.06	32.0
3	MNIST	32	98.96	2.08	1.0
2	MNIST	8	99.06	0.52	4.0
1	MNIST	4	98.81	0.26	8.0
0	MNIST	1	99.06	0.06	32.0

Table 1 Classification Accuracy and Model Compression Results

Table 1 presents classification accuracy and compression metrics for all bitwidth configurations across both datasets. The results show clearly different quantization tolerance between MNIST and CIFAR-10

For MNIST, weight quantization shows negligible impact on accuracy across all bitwidth levels. The 1-bit configuration achieves 99.06% accuracy, slightly exceeding the 32-bit baseline (98.96%), while providing 32 $\times$ model compression. Similarly, 4-bit and 8-bit configurations maintain accuracy within 0.15% of baseline. This suggests that MNIST's high inter-class separability renders weight precision ultimately irrelevant to classification performance.

On the other hand, CIFAR-10 demonstrates clear sensitivity to quantization. The 32-bit baseline achieves 70.45% accuracy, while progressive bitwidth reduction yields systematic degradation: 8-bit (66.81%, -3.64%), 4-bit (63.71%, -6.74%), and 1-bit (62.71%, -7.74%). Something to take note of, 4-bit and 1-bit performance converges near 63%, indicating a quantization floor where further precision reduction yields minimal additional degradation.

The dataset-dependent nature of quantization tolerance is stark: MNIST shows -0.10% degradation at 1-bit while CIFAR-10 shows +7.74% degradation, a 77 $\times$ difference in sensitivity.

This data validates our hypothesis that dataset complexity fundamentally determines acceptable quantization levels.

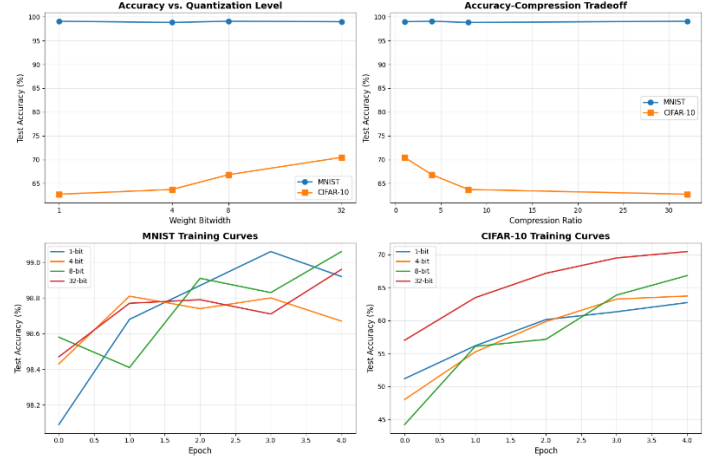


Fig. 1. Quantization performance and tradeoffs. Top left: Test accuracy vs. weight bitwidth for MNIST (blue) and CIFAR-10 (orange). MNIST shows negligible sensitivity while CIFAR-10 exhibits clear degradation. Top right: Accuracy vs. compression ratio, demonstrating non-linear tradeoff for CIFAR-10. Bottom: Training curves showing convergence behavior for each bitwidth configuration over 5 epochs.

Figure 1 (top panels) shows the relationship between quantization level and accuracy. This also shows MNIST maintaining near-constant accuracy across all bitwidths (blue line near 99%), while CIFAR-10 shows monotonic degradation as bitwidth decreases.

The accuracy-compression tradeoff (top-right) reveals non-linear behaviors for CIFAR-10. It moves from 32-bit to 8-bit to incur 3.64% accuracy loss for 4 $\times$ compression. This however, moving from 8-bit to 1-bit incurs only an additional 4.10% loss while achieving 8 $\times$ further compression (0.51% loss per compression factor). This diminishing marginal cost may suggest extreme quantization becomes very efficient at extreme compression ratios.

For MNIST, the tradeoff curve is essentially flat throughout all compression ratios up to 32 $\times$ achieving equivalent accuracy. This may indicate that for simple, well-separated datasets, practitioners can select bitwidth based purely on deployment constraints without sacrificing accuracy.

Quantization at 8-bit weight on CIFAR-10 retains 94% of baseline accuracy (66.81% / 70.45%) while 1-bit retains 89.0%, a 5.8% point difference for 8 $\times$ additional compression. We can conclude that 8-bit is the optimal tradeoff point for complex datasets when accuracy is prioritized, while 1-bit is only viable when extreme compression is paramount and ~10% accuracy loss is acceptable.

The bottom panels of figure 1 presents training curves for all bitwidth configurations. For MNIST, all configurations converge to similar accuracy by epoch 2, with minimal variation thereafter. The 1-bit, 4-bit, and 8-bit

curves overlap, confirming that quantization does not obstruct convergence for simple datasets.

Analysis of the CIFAR-10 training trajectories reveals that while bitwidth reduction impacts final representational capacity, it does not impede, to a concerning degree, optimization convergence. All configurations, ranging from 1-bit to 32-bit, reach stable performance within 3–4 epochs. This suggests that low-precision constraints do not necessitate extended training durations or fundamentally alter learning dynamics. The successful application of the Straight-Through Estimator (STE) ensures reliable gradient propagation even at extreme (1-bit) scales. Consequently, accuracy degradation is attributed to restricted representational capacity rather than stochastic optimization failure. The convergence of 4-bit and 1-bit trajectories toward a shared performance "floor" (~63–64%) suggests a lower bound on accuracy that persists even with varying early-training behaviors. These findings indicate that quantized models exhibit training reliability comparable to full-precision baselines; therefore, deployment performance is primarily a function of bitwidth selection rather than the training procedure itself.

IV. DISCUSSION

Our results demonstrate that quantization thresholds are fundamentally dataset-dependent rather than architecture-dependent. MNIST tolerates extreme 1-bit quantization without accuracy loss (99.06% vs 98.96% baseline), while CIFAR-10 exhibits 7.74% degradation at the same bitwidth. This 77× difference in sensitivity cannot be attributed to architecture, training, or hyperparameters, all were held constant across experiments.

The key differentiator is dataset complexity: MNIST's high inter-class separability requires minimal capacity, while CIFAR-10's low separability demands finer weight precision. This suggests Data Scientists should evaluate quantization strategies based on task difficulty rather than applying universal bitwidth recommendations.

For CIFAR-10, we identified a practical threshold between 4-bit and 8-bit quantization. The 8-bit configuration achieves 94.8% accuracy retention with 4× compression, representing that optimal balance for production deployment. Contrastingly, 4-bit and 1-bit configurations converge near a performance floor of ~63%, suggesting a lower bound on achievable accuracy regardless of further precision reduction.

This study focuses exclusively on weight quantization while maintaining full-precision activations and gradients. While this design enables controlled analysis, real-world deployment often benefits from joint quantization of all network components [14]. Future work should investigate how activation and gradient quantization interact with weight quantization to determine optimal multi-component configurations.

Our experiments used a pretty shallow CNN (~545K parameters) on small-scale datasets (32×32 images). Modern production models contain orders of magnitude more parameters and process higher-resolution inputs. Extending this analysis to large-scale architectures on ImageNet would validate

whether our threshold findings generalize to state-of-the-art systems.

Additionally, we examined only uniform quantization where all layers share the same bitwidth. Mixed-precision quantization, assigning different bitwidths to different layers based on sensitivity analysis, may achieve better accuracy-compression tradeoffs than our constant parameterized approach. Automated search methods for layer-wise bitwidth assignment represent a promising direction for future research.

V. CONCLUSION

This study presented a systematic investigation into the functional relationship between weight quantization bit-depth and neural network performance across divergent levels of stochastic complexity. By isolating weight-specific discretization while maintaining full-precision activations and gradients, we established a controlled empirical framework to characterize the individual impact of bitwidth reduction on classification accuracy. Our results demonstrate that quantization resilience is tied to dataset complexity; low-entropy tasks such as MNIST maintain performance even under extreme 1-bit constraints, whereas high-entropy benchmarks like CIFAR-10 exhibit a distinct performance threshold between 4-bit and 8-bit precision.

The empirical evidence reveals a non-linear compression-accuracy manifold, where the marginal cost of bitwidth reduction diminishes at lower precisions. This is shown empirically by the convergence of CIFAR-10 accuracy toward a representational "floor" near 63% for both 4-bit and 1-bit regimes, a finding that facilitates robust worst-case loss prediction during the deployment planning phase. For complex visual recognition tasks, 8-bit quantization emerged as the optimal Pareto frontier, delivering a 4x reduction in model footprint with less than 4% accuracy, therefore it offers a viable compromise for production environments where efficiency and fidelity are equal priorities. Furthermore, the analysis of training trajectories confirms that quantization does not fundamentally impede optimization dynamics, because all configurations exhibited reliable convergence regardless of numerical precision. This indicates that performance degradation is a consequence of restricted representational capacity rather than optimization failure.

Ultimately, these findings provide a formalized heuristic for practitioners seeking to optimize neural network architectures for resource-constrained environments. While simple tasks enable aggressive 32× compression via 1-bit quantization without performance penalties, complex applications demand a more conservative 8-bit approach to preserve essential predictive accuracy. Future research will extend these inquiries to large-scale architectures and explore the synergistic effects of joint weight-activation-gradient quantization, alongside mixed-precision strategies for layer-wise bitwidth optimization.

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